Positron Emission Tomography Study of Human Brain Functional Development

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From over 100 children studied with 2-deoxy-2[^{18}F]fluoro-D-glucose and positron emission tomography we selected 29 children (aged 5 days to 15.1 years) who had suffered transient neurological events not significantly affecting normal neurodevelopment. These 29 children were reasonably representative of normal children and provided an otherwise unobtainable population in which to study developmental changes in local cerebral metabolic rates for glucose (lCMRGlc). In infants less than 5 weeks old lCMRGlc was highest in sensorimotor cortex, thalamus, brainstem, and cerebellar vermis. By 3 months, lCMRGlc had increased in parietal, temporal, and occipital cortices; basal ganglia; and cerebellar cortex. Frontal and dorsolateral occipital cortical regions displayed a maturational rise in lCMRGlc by approximately 6 to 8 months. Absolute values of lCMRGlc for various grey matter regions were low at birth (13 to 25 μmol/min/100 gm), and rapidly rose to reach adult values (19 to 33 μmol/min/100 gm) by 2 years. lCMRGlc continued to rise until, by 3 to 4 years, it reached values of 49 to 65 μmol/min/100 gm in most regions. These high rates were maintained until approximately 9 years, when they began to decline, and reached adult rates again by the latter part of the second decade. The highest increases of lCMRGlc over adult values occurred in cerebral cortical structures; lesser increases were seen in subcortical structures and in the cerebellum. This time course of lCMRGlc changes matches that describing the process of initial overproduction and subsequent elimination of excessive neurons, synapses, and dendritic spines known to occur in the developing brain. The determination of changing metabolic patterns accompanying normal brain development is a necessary prelude to the study of abnormal brain development with positron emission tomography.

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During development, the brain undergoes the sequential anatomical, functional, and organizational changes necessary to support the complex adaptive behavior of a fully mature normal individual. The delineation of developmental changes occurring in different brain regions would provide a means of relating various behavioral phenomena to maturation of specific brain structures, thereby enhancing our understanding of structure–function relationships in both normal and disease states.

One approach to the study of these relationships has been to measure regional substrate utilization at different stages of cerebral maturation. Since the principal brain substrates for energy production are glucose and oxygen, the determination of their regional alterations would provide a measure of the local energy requirement for maintenance processes and functional activity. This approach has been used in the study of brain–behavior relationships in various animal models [1, 18, 31], and has generated important concepts in our understanding of brain function. Alternatively, because of the close relationship between local cerebral blood flow (lCBF) and local cerebral metabolic rate under normal conditions [54], an indirect assessment of local functional activity can also be obtained by the measurement of lCBF during development [4, 29, 30, 43].

The development of positron emission tomography (PET) [47], which employs tracer kinetic measurements of compounds labelled with positron-emitting isotopes [48, 49], has enabled us to apply directly the regional substrate utilization approach in studying human brain development. Using 2-deoxy-2[^{18}F]fluoro-D-glucose (FDG) and PET, we have measured local cerebral metabolic rates for glucose (lCMRGlc) in infants and children during postnatal brain development.

In an earlier communication [6], we used relative measures of lCMRGlc to determine the temporal relationship of human functional development during the first 18 months of life. These studies demonstrated that the order of functional development, from the relatively earlier maturation of phylogenetically older brain structures to the later maturation of newer structures, is in general agreement with behavioral, neuro-

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Table 1. Patient Summary

Age, Sex	Clinical Information	Laboratory Data	Medication on Day of Study	Period of Follow-Up after Onset of Neurological Event
5 days, M	42 weeks gestation; mild fetal distress; Apgar scores 2 at 1 minute, 8 at 5 minutes; seizures on days 1 and 2	EEG multifocal spikes, CT scan normal	Phenobarbital	24 months
5 weeks, M	Tonic seizures at age 4 weeks	EEG normal, CT scan normal	Phenobarbital	16 months
8 weeks, F	Onset of myoclonic jerks at age 2 weeks	EEG normal, CT scan normal	None	4 months
13.5 weeks, F	Asymptomatic, left facial port-wine nevus (? Sturge-Weber syndrome)	EEG normal, CT scan normal	None	22 months
4 months, M	Normal birth; one seizure on day 1, none since	EEG normal, CT scan normal	Phenobarbital	17 months
6.25 months, F	Onset of seizures at age 5 months	EEG normal, CT and MRI scans normal	Phenobarbital	3 months
7.6 months, M	Opsoclonus and myoclonus developed at age 6.5 months, responded well to adrenocorticotropic hormone	EEG paroxysmal generalized spike and wave activity, CT scan normal	Phenobarbital, clonazepam	22 months
1 year, F	Asymptomatic, left facial port-wine nevus (? Sturge-Weber syndrome)	EEG normal, CT scan normal	None	32 months
1.5 years, F	Glaucoma OS, bilateral facial port-wine nevi (? Sturge-Weber syndrome)	EEG normal	None	36 months
1.5 years, M	Glaucoma OD, right facial port-wine nevus (? Sturge-Weber syndrome)	EEG normal, CT scan normal	None	27 months
1.6 years, M	Onset of apneic episodes at age 9 months	EEG normal, CT scan normal	None	18 months
3.5 years, F	Onset of seizures at age 1.5 years	EEG normal, CT scan normal	Carbamazepine	26 months
3.7 years, F	Complex febrile seizures	EEG normal	Phenobarbital	43 months
5.7 years, F	Onset of rolandic epilepsy at age 2.5 years	EEG right centro-temporal spike-wave discharges, CT scan normal	Carbamazepine	4 years
6.7 years, M	Episodic generalized paresthesias (? seizures)	EEG normal, CT and MRI scans normal	None	11 months
6.8 years, M	Visual hallucinations beginning at age 5.5 years	EEG normal, CT and MRI scans normal	Carbamazepine, clorazepate	18 months
7 years, F	2 grand mal seizures at age 5 years	EEG normal, CT scan normal	Carbamazepine	34 months
7.1 years, F	A single nocturnal seizure at age 7 years	EEG infrequent right centrotemporal spike-wave discharges, CT scan normal	Carbamazepine	5 months
7.2 years, M	15 episodes of tunnel vision at age 7 years	EEG normal, CT and MRI scans normal	None	18 months
7.5 years, F	Onset of complex partial seizures at age 6 years	EEG occasional generalized spike-wave discharges, CT and MRI scans normal	Phenobarbital carbamazepine	24 months
8.4 years, M	Rolandic seizures beginning at age 5 years	EEG left centrotemporal spikes, CT scan normal	Phenobarbital	4 years

Table 1. (Continued)

Age, Sex	Clinical Information	Laboratory Data	Medication on Day of Study	Period of Follow-Up after Onset of Neurological Event
8.5 years, M	Hypoplastic kidneys, on chronic ambulatory peritoneal dialysis; febrile seizures in infancy (? myoclonic seizures at age 6 years)	EEG generalized irregular spike-wave activity only during sleep, CT scan normal	Clonazepam	32 months
8.6 years, M	Hemolytic-uremic syndrome at age 8 years with convulsion in acute phase	EEG normal, CT scan normal	None	9 months
9.3 years, M	Onset of complex partial seizures at age 9 months, infrequent episodes	EEG normal, CT scan normal	Carbamazepine	7 months
9.5 years, F	Asymptomatic, left facial port-wine nevus (? Sturge-Weber syndrome)	EEG normal, CT scan normal	None	10 years
9.8 years, M	Onset of left simple partial motor seizures at age 3 years	EEG right paracentral spikes, CT and MRI scans normal	Carbamazepine, phenytoin	6 years
12 years, F	Onset of complex partial seizures at age 2 years, infrequent episodes	EEG normal, CT scan normal	Carbamazepine	10 years
13.5 years, M	Onset of complex partial seizures at age 10 years, infrequent episodes	EEG normal, CT scan normal	Carbamazepine	3 years
15.1 years, M	Onset of seizures with migraine at age 11 years	EEG normal, CT scan normal	None	4 years

M = male; F = female; EEG = electroencephalogram; CT = computed tomography; MRI = magnetic resonance imaging; OS = left eye; OD = right eye.

physiological, and anatomical alterations known to accompany infant development. We now report absolute values of lCMRGlc for different brain regions from birth to adolescence and compare them with normal adult values.

Materials and Methods

Patient Population

We have retrospectively selected 29 subjects from over 100 infants and children studied with FDG-PET at UCLA School of Medicine since 1983. Clinical summaries of these 29 children are provided in Table 1. Of the 29 individuals, 24 were selected from our research protocol designed to evaluate the role of PET in seizures and epilepsy. Each had suffered transient neurological events not significantly altering neurological development on close follow-up. Informed consent was obtained in all cases. The remaining 5 subjects were asymptomatic, but because they had facial capillary nevi they were evaluated with FDG-PET for possible Sturge-Weber syndrome. The parents of each of these 5 subjects were anxious to accept any noninvasive procedure with potential diagnostic value. FDG-PET did not disclose any hemispheric asymmetry of lCMRGlc in any one of these 5 patients. We believe that these 29 children are reasonably representative of normal children and provide an opportunity to measure lCMRGlc changes during normal brain development, since entirely normal children cannot be studied with PET for ethical reasons. Indeed, the older children in this subpopulation were all attending school and performing well in classes for normal children. Of the infants studied, none had been born prematurely and all continue to develop normally. There were 16 males and 13 females, and their ages ranged from 5 days to 15.1 years at the time of study. Although 18 of the 29 subjects were taking daily anticonvulsants at the time of PET, none displayed sedation or other side effects.

PET Procedure

All studies were performed in accordance with the policies of the UCLA Human Subject Protection Committee. Subjects were fasted (except for water) for 4 hours prior to PET. One hour prior to the study, golden disk electrodes were applied to the patient's scalp for later electroencephalographic (EEG) monitoring. Then a venous catheter was inserted in a hand vein for blood collection. The blood was arterialized by placing the hand into a water chamber heated to 44°C [48]. Radial arterial catheters were used in 3 subjects.

FDG (0.143 mCi/kg) was administered intravenously as a pulse at a different site from the catheter, and serial blood samples were collected to determine plasma FDG and glucose concentrations as described elsewhere [48]. In infants younger than 2 years, microtechniques were employed to minimize total blood collected (< 5 ml). The dose of FDG was approximately 25% lower than that used in adults, and in children results in a whole body radiation exposure of approximately 300 mrads. The dose to the brain is about 500 mrads (as compared to 1 to 3 rads in x-ray computed tomography).

EEG activity was monitored, and the children were closely observed for seizure activity. The EEG recordings were also useful in the early detection of drowsiness in some children. When a child became drowsy, as evidenced by diminishing amplitude and slowing of EEG activity, he or she was gently

tapped on the shoulder to be kept awake and alert. Other than this occasional stimulus, all visual, auditory, and other sensory stimuli were minimized by dimming the lights and discouraging talking and other forms of interaction. Patients' eyes and ears were open during the study.

Forty minutes after FDG injection scanning of the brain was initiated using the NeuroECAT positron tomograph (CTI, Knoxville, TN) with a spatial resolution of 8.4 mm in the plane of section and a 12.4 mm slice thickness [19]. A head holder minimized head movement during scanning [38]. To confirm true axial positioning, 2 rectilinear images (anteroposterior and lateral views) were obtained prior to generating a standard set of 12 tomographic images parallel to the canthomeatal line.

Data Analysis

Patterns of lCMRGlc on the PET images were initially assessed by inspection. Then the images were displayed on a display monitor, and regions of interest were drawn for different brain structures using anatomical atlases, patients' CT scans, and standard lCMRGlc functional maps as references [39]. Using the operational equation of Sokoloff and associates [55] as modified by Phelps and associates [48] and Huang and associates [21], lCMRGlc for the different brain regions was then calculated and expressed in μmol/min/100 gm. Whenever a brain region appeared on more than 1 tomographic level, the lCMRGlc value used for that region was a bilateral average (weighted by area) of the values for each tomographic plane where that region appeared. The lumped constant and rate constants employed were those measured previously for normal young adults [21, 48].

The anatomical distribution and absolute values of lCMRGlc in the children were compared to those of adults studied under similar conditions. The latter group consisted of 7 young healthy volunteers (aged 19 to 30 years; mean age 24.4 years; 5 males, 2 females) whose detailed neurological and psychological examinations disclosed no abnormalities.

Results

The PET images revealed several distinct patterns of lCMRGlc in the first year of life (Fig 1) as described below. Although the anatomical distribution of lCMRGlc by 1 year of age was similar to that of adults, absolute values of lCMRGlc were lower than adult rates (Fig 2 and Table 2). Adult rates were reached by approximately 2 years of age; however, lCMRGlc for all regions measured continued to increase, and by 3 to 4 years attained values that exceeded those of adults by a factor of approximately 2 (see Table 2). These high values of lCMRGlc were maintained until approximately 9 years of age, when they began to decline, and reached adult rates again by the end of the second decade. Although this general pattern of maturational rise and decline of lCMRGlc was seen in all brain regions studied, there were individual differences in the rates of maturation among structures as described below.

Statistical analysis of these data consisted of multiple *t*-tests between age groups for each structure using the Bonferroni adjustment for multiple testing. An acceptance criterion of $p < 0.05$ or better was employed for statistical significance. Although mean lCMRGlc values at 1 to 2 years were higher than those from birth to 1 year for all structures (see Table 2), the differences were not statistically significant at the 5% level except for frontal cortex because of the use of multiple testing adjustment. Differences between the birth to 1 year and 3 to 8 years groups were statistically significant for all structures measured ($p < 0.01$). Differences between the 3 to 8 years and 9 to 15 years groups were statistically significant ($p < 0.05$) for all structures except the caudate and lenticular nuclei, brainstem, and cerebellum. Comparisons with the adult group were significantly different for the 3 to 8 and 9 to 15 years age groups, except for brainstem and cerebellum (cerebellum showed a significant difference for the 3 to 8 years age group only). Finally, comparisons between adult and 1 to 2 years age groups were not statistically significant throughout.

Cerebral Cortex

In the neonatal period (less than 4 weeks of age), the most prominent area of metabolic activity in the cerebral cortex was the primary sensorimotor area (see Fig 1A). The remaining cerebral cortical regions had comparatively lower lCMRGlc. By approximately 4 weeks, lCMRGlc in parietal cortex had increased only slightly (see Fig 1B), and by approximately the second postnatal month, lCMRGlc had slightly increased in calcarine and temporal cortices also (see Fig 1C). Considerable maturational rises in lCMRGlc were seen in the anterior parietal, temporal, and calcarine cortices by approximately 3 months postnatally (see Fig 1D). However, most of the frontal cortex, as well as dorsolateral occipital (visual association) cortical regions, was slower to display a developmental increase. Although lCMRGlc in frontal cortex, particularly the lateral prefrontal regions, had increased slightly by 6 months (see Fig 1E), both frontal and visual-association cortical lCMRGlc continued to rise considerably during the next several months (see Fig 1F, G). By 1 year, the lCMRGlc pattern resembled that in adults (see Fig 1G, H).

As depicted in Table 2, mean lCMRGlc for cerebral cortical regions in the first year of life ranged from 65 to 86% of the corresponding adult values. The lCMRGlc for the sensorimotor cortex, which was the earliest cortical region to display its maturational rise, was the closest to the adult rates during the first year compared to other cortical structures. After the first year, the maturational curves for all cerebral cortical regions displayed a similar pattern (see Fig 2A, B). Between 3 to 8 years, mean lCMRGlc for most cerebral cortical regions were 190 to 226% of corresponding adult values. As a whole, cerebral cortical

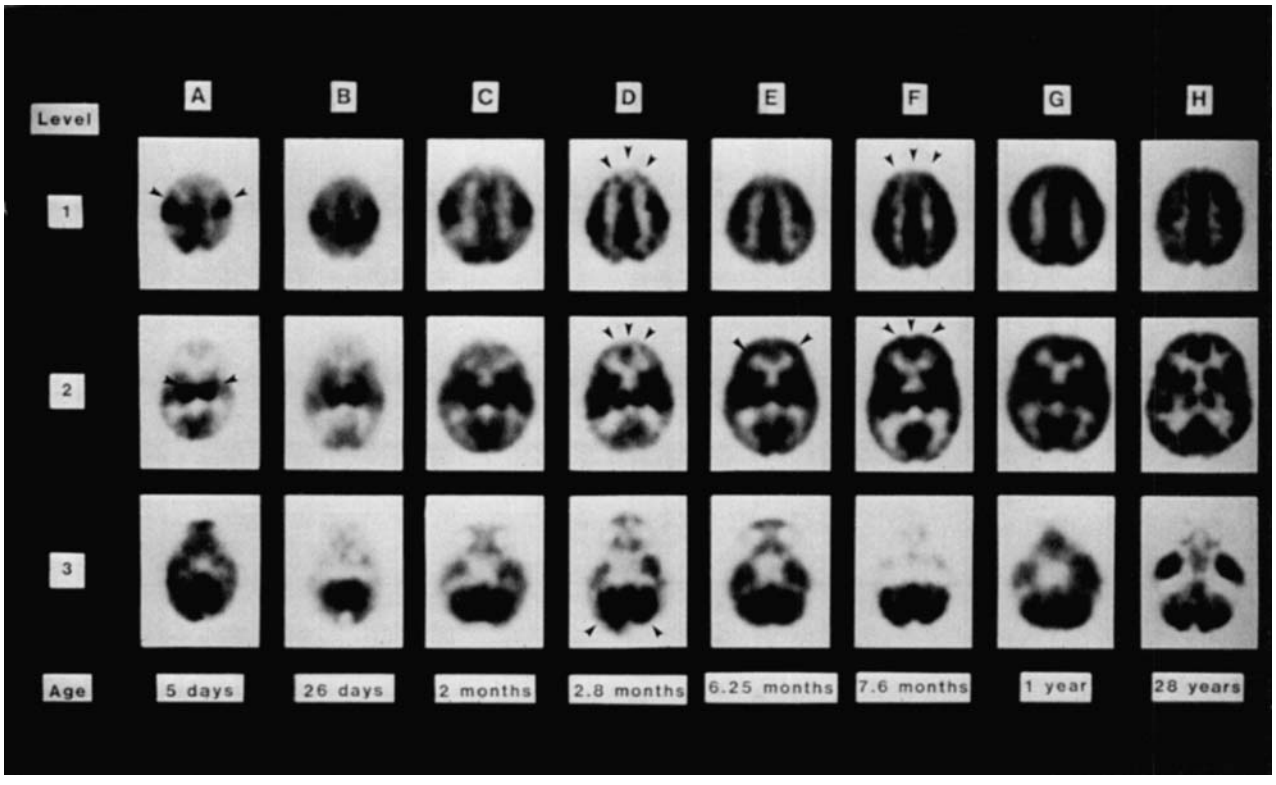


Fig 1. 2-deoxy-2[^{18}F]fluoro-D-glucose (FDG)-positron emission tomography (PET) images illustrating developmental changes in local cerebral metabolic rates for glucose (lCMRGlc) in the normal human infant with increasing age, as compared to lCMRGlc of the adult (image sizes not on same scale). Level 1 is a superior section, at the level of the cingulate gyrus. Level 2 is more inferior, at the level of caudate, putamen, and thalamus. Level 3 is an inferior section of the brain, at the level of cerebellum and inferior portion of the temporal lobes. Grey scale is proportional to lCMRGlc, with black being highest. Images from all subjects are not shown on the same absolute grey scale of lCMRGlc; instead, images of each subject are shown with the full grey scale to maximize grey scale display of lCMRGlc at each age. Changes in absolute lCMRGlc with age appear in Figure 2 and Table 2. In each image, the anterior portion of the brain is at the top of the image and the left side of the brain is at the left of the image. (A) In the 5-day-old, lCMRGlc is highest in sensorimotor cortex, thalamus, cerebellar vermis (arrows), *and brainstem (not shown). (B, C, D) lCMRGlc gradually increases in parietal, temporal, and calcarine cortices; basal ganglia; and cerebellar cortex* (arrows), *particularly during the second and third months. (E) In the frontal cortex, lCMRGlc increases first in the lateral prefrontal regions by approximately 6 months. (F) By approximately 8 months, lCMRGlc also increases in the medial aspects of the frontal cortex* (arrows), *as well as the dorsolateral occipital cortex. (G) By 1 year, the lCMRGlc pattern resembles that of adults (H).*

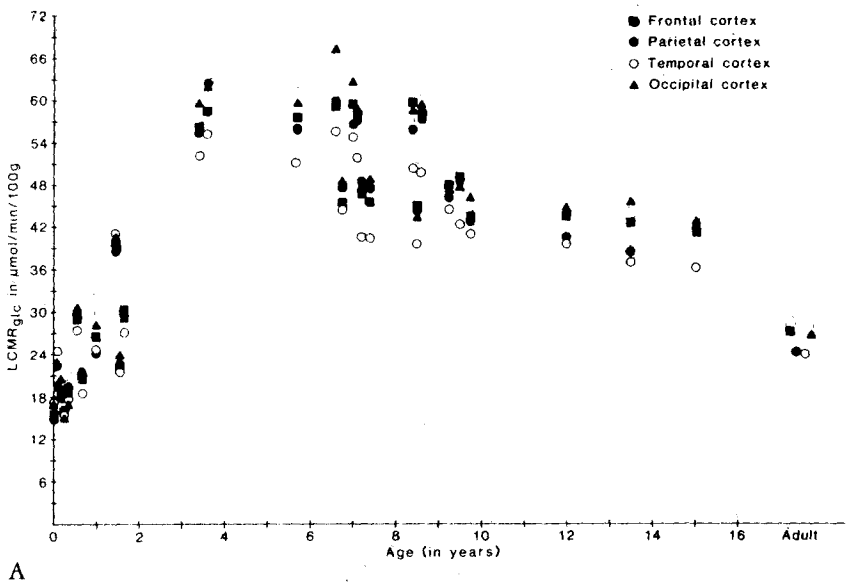
structures underwent the highest proportional increase in lCMRGlc over their mature rates when compared to other portions of the brain.

Basal Ganglia and Thalamus

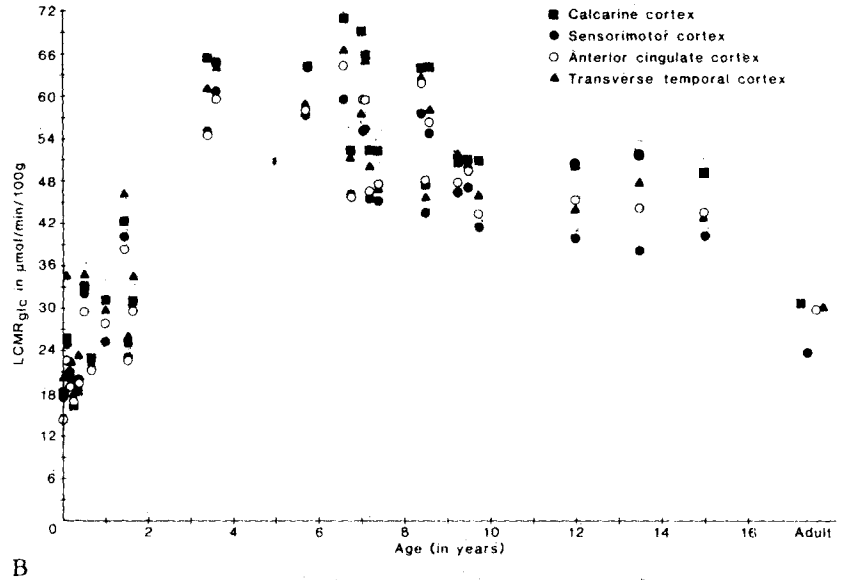
The basal ganglia and thalamus displayed developmental increases in lCMRGlc earlier than most areas of the cerebral cortex (see Figs 1A–C, 2C). The thalamus was one of the earliest structures to show a maturational rise in lCMRGlc and was prominently active even in the 5-day-old infant (see Fig 1A). Indeed, mean lCMRGlc for the thalamus was already 84% of its adult rates during the first year; for lenticular nuclei it was 75% and for caudate nuclei 69%. Although peak values of lCMRGlc attained by the basal ganglia and thalamus during childhood exceeded those of corresponding structures in the adult, the magnitude of increase (155 to 162%) was less than that of cerebral cortex (see Table 2).

Cerebellum

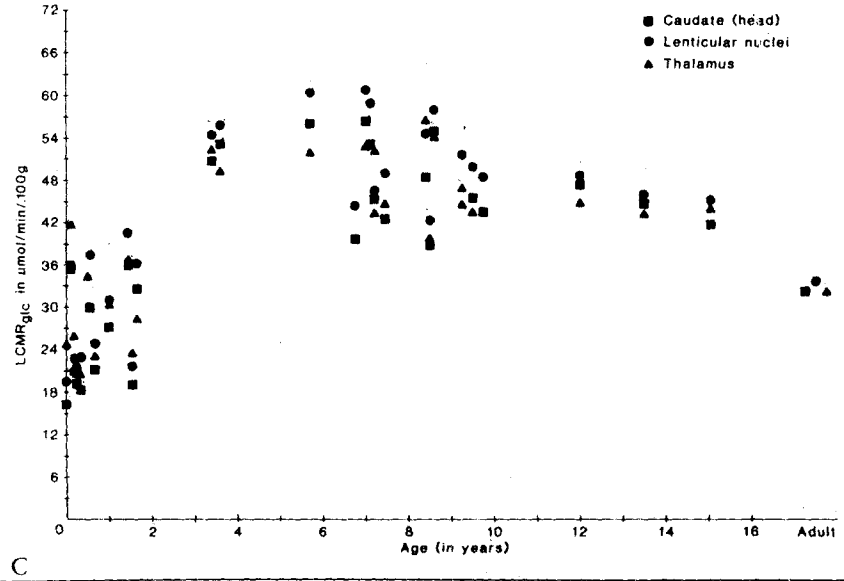
During the first year, lCMRGlc for the cerebellum as a whole was closer to adult rates than any other portion of the brain (see Table 2). Within the cerebellum, however, a heterogeneous pattern of maturation was observed. The centrally located, phylogenetically older portion (vermis) of the cerebellum was the first to display a maturational rise (see Fig 1A, B), showing relatively high lCMRGlc as early as 5 days of age. The cerebellar hemispheres, or neocerebellum, had com-


Frontal cortex
Parietal cortex
Temporal cortex
Occipital cortex
LCMRglc in µmol/min/100g
Age (in years)
Adult

A


Calcarine cortex
Sensorimotor cortex
Anterior cingulate cortex
Transverse temporal cortex
LCMRglc in µmol/min/100g
Age (in years)
Adult

B


Caudate (head)
Lenticular nuclei
Thalamus
LCMRglc in µmol/min/100g
Age (in years)
Adult

C

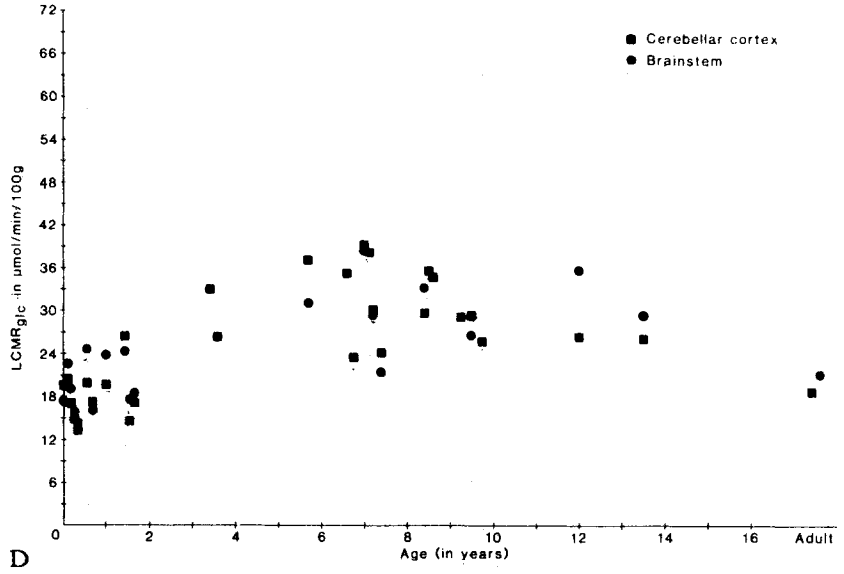


Fig 2. Absolute values of local cerebral metabolic rates for glucose (lCMRGlc) for selected brain regions plotted as a function of age for all 29 infants and children, and corresponding adult values. In the infants and children, points represent individual values of lCMRGlc; in adults, points are mean values from 7 subjects, in which the size of the symbols equals the standard error of the mean (also see Table 2). (A, B) Selected regions of cerebral cortex; (C) basal ganglia and thalamus; (D) cerebellar cortex and brainstem.

Table 2. Local Cerebral Metabolic Rates for Glucose[a] (lCMRGlc) in µmol/min/100 gm and as a Ratio[b] to Adult Rates for Selected Brain Regions During Development

	Birth–1 year (n = 7)		1–2 years (n = 4)		3–8 years (n = 12)		9–15 years (n = 6)		Adult 19–30 years (n = 7)
Structure	lCMRGlc	Ratio	lCMRGlc	Ratio	lCMRGlc	Ratio	lCMRGlc	Ratio	lCMRGlc
Cerebral hemisphere	20.35 ± 1.81	0.84	27.36 ± 3.23	1.13	48.07 ± 1.68	1.98	39.23 ± 0.79	1.62	24.27 ± 1.09
Cerebral cortex									
Frontal cortex	19.68 ± 1.83	0.73	29.65 ± 3.65	1.09	54.01 ± 1.82	1.99	44.59 ± 1.21	1.64	27.11 ± 1.26
Parietal cortex	20.23 ± 1.90	0.84	28.76 ± 3.54	1.19	53.98 ± 1.66	2.24	43.10 ± 1.50	1.78	24.15 ± 1.40
Temporal cortex	20.00 ± 1.60	0.83	28.64 ± 4.27	1.19	48.94 ± 1.69	2.04	40.05 ± 1.31	1.67	23.98 ± 1.15
Occipital cortex	20.54 ± 1.92	0.76	30.39 ± 3.50	1.13	56.35 ± 2.11	2.10	45.66 ± 0.78	1.70	26.85 ± 1.09
Calcarine cortex	22.07 ± 2.20	0.72	32.39 ± 3.55	1.06	61.01 ± 2.22	1.99	50.45 ± 0.35	1.65	30.59 ± 1.15
Sensorimotor cortex	22.17 ± 1.93	0.93	29.73 ± 3.72	1.25	52.85 ± 1.75	2.22	42.04 ± 1.52	1.76	23.83 ± 1.30
Anterior cingulate cortex	20.53 ± 1.85	0.69	29.61 ± 3.20	0.99	55.08 ± 1.88	1.85	45.36 ± 1.01	1.52	29.79 ± 1.31
Transverse temporal cortex	25.47 ± 2.44	0.86	34.00 ± 4.38	1.14	57.25 ± 2.10	1.92	47.00 ± 1.41	1.57	29.86 ± 1.33
Basal ganglia and thalamus									
Caudate (head)	22.95 ± 2.67	0.71	28.77 ± 3.67	0.89	48.86 ± 1.96	1.52	44.76 ± 0.83	1.39	32.20 ± 1.34
Lenticular nuclei	26.42 ± 2.70	0.79	32.36 ± 4.02	0.97	53.08 ± 1.98	1.58	48.15 ± 0.96	1.44	33.50 ± 1.27
Thalamus	27.15 ± 2.97	0.84	29.48 ± 2.95	0.92	49.54 ± 1.71	1.54	43.81 ± 0.28	1.36	32.19 ± 1.96
Cerebellum									
Cerebellar cortex	17.59 ± 1.02	0.93	19.50 ± 2.50	1.03	32.28 ± 1.56	1.71	27.38 ± 0.75	1.45	18.85 ± 0.86
Brainstem	18.63 ± 1.44	0.89	21.17 ± 1.80	1.01	30.74 ± 2.75	1.46	30.61 ± 2.68	1.45	21.04 ± 1.12

[a]Mean ± SEM.
[b]Ratio to mean adult lCMRGlc.

paratively lower lCMRGlc in the neonatal period, and only displayed a maturational rise in lCMRGlc by the third postnatal month (see Fig 1D). During childhood, peak values of lCMRGlc in the cerebellum were approximately 175% of adult rates for this structure (see Table 2).

Brainstem

The maturational pattern of the brainstem was similar to that of the cerebellum in that rates were relatively advanced at birth. During the first year, mean lCMRGlc for brainstem was approximately 82% of mature rates, and subsequent peak values were ap-

proximately 170% of corresponding adult values (see Table 2).

Discussion

Methodological Issues

Although approximately two thirds of the subjects in this study were taking anticonvulsants daily, sedative effects were not seen. Nevertheless, in PET studies of adult intractable epileptics taking multiple anticonvulsants, lCMRGlc in cerebral cortex increased by a mean of 37% after withdrawal of phenobarbital or primidone [58], and a mean of 13% after withdrawal of phenytoin [57]. It seems unlikely that our findings were significantly affected by anticonvulsant use, since subjects of all ages were taking drugs at the time of PET and drug effects on lCMRGlc, if they occurred, would tend to be random.

A second methodological issue in our study involves the use of rate constants from normal young adults for calculating lCMRGlc in children. This has relatively little impact on the absolute values of lCMRGlc, because the rate constants appear with terms in the equation that contain exponentials to a negative power multiplied by time. Thus, at the late times after measurement these factors are small, and variations in the exact values of the rate constants cause only small errors over the range of calculated lCMRGlc values in this study [21, 22, 48, 55].

Earlier Studies of Cerebral Blood Flow and Substrate Utilization During Development

Prior to the development of PET, there had been no adequate noninvasive method to study changes of local substrate utilization in the developing human brain. Nevertheless, in their studies of cerebral blood flow and oxygen utilization, Kennedy and Sokoloff [32] demonstrated that the average global cerebral blood flow in 9 normal children (aged 3 to 11 years) was approximately 1.8 times that of normal young adults. Similarly, average cerebral oxygen utilization was approximately 1.3 times higher in children than in adults.

The subsequent development of autoradiographic techniques for measuring lCBF [33, 52] and lCMRGlc [55] in laboratory animals enabled substrate utilization to be determined in individual anatomical structures during cerebral maturation. These previous investigations have suggested that during different stages of development cerebral structures with relatively high lCBF or lCMRGlc in general determine the predominant behavioral pattern at the particular stage [1, 4, 29–31, 43]. For example, using [^{14}C]2-deoxyglucose autoradiography, Kennedy and associates [31] found lCMRGlc in neonatal monkeys to be lower than adult rates in structures above the midbrain, particularly striate and the inferior temporal cortex. In contrast, lCMRGlc in neonatal auditory and somatosensory cortical areas were similar to mature rates, and neonatal thalamic rates were also relatively high when compared to other structures. These findings supported the concept that, in general, a rise in metabolic rate of a particular structure marks the time of its contribution to the animal's behavior. In the case of the monkey, structures with the highest lCMRGlc in the neonatal period dominated the behavior at that age [31].

Correlation Between lCMRGlc and Behavior During Human Development

Our data on the patterns of lCMRGlc in human neonates are congruous with the thesis that there is a relationship between a metabolic increase within neuroanatomical structures and the emergence of corresponding function. Neonatal behavior is primarily dominated by subcortical brain structure activity. Intrinsic brainstem reflexes, such as the Moro, root, and grasp reflexes, are prominent. Visuomotor function is present only in rudimentary form [60], and cortical function is mostly limited to primary sensory and motor areas. The prominent metabolic activity in sensorimotor cortex (see Fig 1A) is also consistent with its relatively early morphological maturation compared to other cortical areas [51].

An interesting finding in the neonatal PET images is the relatively low lCMRGlc in the striatum compared to the more functionally mature thalamus (see Fig 1A, B). This lCMRGlc relationship is also seen in adults with Huntington's disease [35]. Since normal newborn infants manifest nonpurposeful limb movements similar to those seen in Huntington's disease, the mechanism of chorea may be related to a functional imbalance in the interaction among striatum, thalamus, and cerebral cortex. Pathological chorea may result from striatal loss of function, whereas newborn physiological chorea occurs prior to emergence of significant lCMRGlc in the striatum and much of the cerebral cortex. It would, therefore, be of interest to determine whether a similar relationship in lCMRGlc among these structures exists in other clinical entities where chorea is a prominent feature.

By approximately 3 months, when striatal lCMRGlc has approached that of thalamus (see Fig 1D), nonpurposeful limb movements have been replaced by more coordinated reaching movements. During this time, lCMRGlc has also increased in structures important for visuospatial and visuo-sensorimotor integration, such as parietal cortex, primary visual (calcarine) cortex, and cerebellar hemispheres. Indeed, infants between 2 and 3 months of age often open their hands during forward extension of the arm while visually fixing on an object as a preparatory maneuver to manipulate the object [59]. The increasingly sophisticated nature of these reaching movements is consistent with the notion that, whereas visual function in newborns

may be mediated by phylogenetically older (subcortical) visual structures, the primary visual (calcarine) cortex begins to play a more important role in visual function by 2 to 3 months [3]. The lCMRGlc in the visual association areas, however, remains relatively low during this period, a finding consistent with the delayed anatomical maturation of the dorsolateral occipital cortex compared to calcarine cortex [51].

Another behavioral hallmark of 2- to 4-month-old infants is the attenuation and gradual disappearance of intrinsic brainsteam reflexes, presumably as a result of increasing cortical influence [2, 45]. Since the frontal cortex remains relatively hypometabolic compared to parietal, temporal, and calcarine cortices during this time (see Fig 1C, D), cortical suppression of brainstem reflexes (assuming this to be the mechanism) is probably not exclusively the result of frontal cortical input. These primitive newborn reflexes are also often seen in demented individuals suffering from Alzheimer's disease [46], where PET has demonstrated the early decrease of parietal, prior to frontal lobe, metabolism [9, 15, 34]. Therefore, the concept of *frontal release signs*, a phrase often used to denote the emergence of primitive reflexes in both demented and normal elderly individuals, may be more accurately referred to as *cortical release signs.*

The EEG, another measure of cerebral cortical activity, also undergoes considerable maturation during the second and third postnatal months. During this time, newborn EEG patterns, such as trace alternant, frontal rhythmic delta, and frontal sharp transients, disappear, and the precursors of alpha rhythm appear [28]. Thus, it is not surprising that increases of lCMRGlc in cerebral cortical structures should occur during this period of cortical maturation as evidenced by dramatic EEG changes.

By 8 months postnatally, lCMRGlc had increased in dorsolateral occipital cortex and much of the frontal cortex (see Fig 1F). These increases coincide with the appearance of higher cortical and cognitive function [27]. The infant now shows a more sophisticated interaction with his or her surroundings and exhibits the phenomenon of stranger anxiety [12]. In addition, the infant improves his or her performance on the delayed response task, a commonly used neuropsychological paradigm for evaluating prefrontal lobe integrity [16, 17]. Together, these changes in the skills of the infant imply increasing function in the prefrontal cortex. Neuroanatomically, this stage is accompanied by an expansion of dendritic fields [53] and an increase in capillary density [8] of human frontal cortex.

Possible Causes of High lCMRGlc During Development

Although by 8 to 12 months of age the anatomical distribution of lCMRGlc seen with PET qualitatively resembled that of young adults, and by 2 years absolute lCMRGlc for most structures were similar to those of adults, brain development was far from complete. In virtually all brain regions measured, lCMRGlc continued to increase, although to varying degrees depending upon the specific structures, reaching very high values compared to adults. Only during the second decade did lCMRGlc decline to approximate adult values (see Fig 2). Evidence in support of these high metabolic rates during childhood was first reported by Kennedy and Sokoloff [32] some 30 years ago in a study alluded to earlier in this discussion.

There are a number of possible explanations for the high energy demands of the brain during development. First, there is now ample evidence in humans [51] and in other species [10, 36, 44] that during development the brain produces a vast excess of neurons, synapses, and dendritic spines. For example, the phenomenon of polyneuronal innervation, where synaptic targets receive innervation from more neurons during development than will remain in the adult, has been well documented (reviewed in [50]). The overproduction of neurons and their synaptic contacts is biologically advantageous in reducing the genetic load that would otherwise be required for specifically programming the enormous numbers of synaptic contacts in the nervous system [5, 25]. Many neurons subsequently die, and there is a regression of dendritic spines and synapses [7]. This form of cell death occurs early in development—by 2 years of age in humans [51]—but the loss of synaptic elements is a more protracted process. For example, the concentration of synaptic contacts in both human frontal [23] and visual [24] cortices of children up to 11 years old exceeds those of corresponding regions in adults. Specifically, at age 7 years, when the child's brain is almost identical in size and weight to that of the adult, average synaptic density in frontal cortex is about 1.4 times the adult value [23].

The biological rules governing regression of neuroanatomical elements are poorly understood, but these phenomena are believed to account, at least in part, for nervous system plasticity [11]. The concept that there are mechanisms that act to retain those pathways in which patterns of external stimuli induce activity and eliminate potential connections not so activated has been termed *functional validation* by Jacobson [25], and *selective stabilization* by Changeux and Danchin [5]. This process would account for the contribution of early experience of the developing individual toward the final neuroanatomical composition and neurophysiological representation within the nervous system. This notion is supported by data from animal studies [13, 40, 41, 56, 61].

Our finding of higher lCMRGlc in children compared to adults is consistent with the excessive numbers of dendritic processes and synapses in childhood,

since these elements account for most of the glucose utilized [26, 37, 42]. Therefore, the large surface area of an excess number of processes might lead to high resting glucose metabolic rates for maintenance of membrane potentials. As the density of processes diminishes, lCMRGlc also decreases to approach adult levels.

A second cause for high lCMRGlc in the developing brain may be excessive fuel expenditure by oligodendroglia during myelination, which continues, in the human brain, throughout the first decade of life [20, 62], and undergoes remodelling throughout life [14]. Conversely, incomplete myelination of brain pathways may result in suboptimal conduction efficiency, thus requiring greater energy expenditure. The relative contributions of these possible mechanisms in accounting for the high lCMRGlc in the developing brain can only be speculated. Since myelin remodelling is a slow process, it is unlikely to contribute significantly to immediate energy expenditure in the brain. Nevertheless, it is possible that all of these factors, and perhaps others, collectively account for the high lCMRGlc in the developing brain.

Conclusion

Our findings support the commonly accepted view that brain maturation in humans proceeds at least into the second decade of life. This study illustrates the potentially powerful approach PET provides in the study of human brain development. PET has opened a new window on our understanding of neuroanatomical correlation with brain function and behavior. Since both the anatomical distribution and absolute values of lCMRGlc in children are age dependent, the precise mapping of these maturational changes in the normally developing brain is a prerequisite to PET studies of abnormal brain development in children ranging from various learning disabilities to severe psychomotor retardation. Finally, by establishing changes in lCMRGlc during normal development, it will be possible to use PET in the study of central nervous system plasticity in children with early brain injury and other disease processes.

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